

GOLD PRICE PREDICTION USING MACHINE LEARNING

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ABSTRACT

The objective of this research is to determine a machine learning model that can effectively predict gold prices (GLD) from financial indicators, such as the Silver Trust (SLV), S&P 500 index (SPX), the Euro to US Dollar exchange rate (EUR/USD) and US Oil Fund (USO). We build a dataset based on past data from Investing.com, between January 2015 to February 2025. The target variable was used as gold prices (GLD), and SPX, USO, SLV, and EUR/USD were utilized as the main influencing factors. Several supervised machine learning models, like Random Forest, Decision Trees, Linear Regression, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN), were used to establish intricate (visualize) patterns within financial data. The performance of the models was measured through Mean Absolute Error (MAE) and Root Mean Square Error (RMSE). The results show that the Random Forest Regressor is the best model for predicting gold prices; hence, it can be used as a helpful tool for analysts and investors.

Keywords: Gold Price Prediction, Random Forest, Decision Tree, Linear regression, K-Nearest Neighbours (KNN), EUR/USD (Exchange Rate).

I. INTRODUCTION

Gold has been a significant metal for centuries and it is considered a precious resource. Gold is not only a valuable metal. It is an essential element in cultures, economies, and individual investments. Gold is among the significant financial assets for countries like India, China, Indonesia, and many more. Gold is considered the ideal jewel. To keep their currencies stable, countries hold gold reserves as protection against inflation and financial crises and provide economic stability. It goes up and down in value based on the demand in the market and economic conditions. Unlike other assets, gold price volatility affects traders, investors, and financial institutions. Accurate forecasting of gold prices allows stakeholders to make financial decisions. This study seeks to create a model that can predict gold prices according to past trends. The model's purpose is to help investors and traders by the recognition of trends and patterns in gold prices. The data has been used in this study ranges from January 2015 to February 2025. To gain insight into the data set, we employ some exploratory data analysis (EDA) methods. We used machine learning models like Random Forests, Decision Trees, Linear Regression, Support Vector Machines (SVM), and K-nearest neighbours. This paper presents the development of a gold price prediction model.

II. LITERATURE REVIEW

In this research, they predicted gold prices by using economic variables such as the stock profit exchange, silver price, EUR/USD, and United States oil ETF. They used machine learning algorithms such as multiple linear regression, random forest, and gradient boosting to predict the price of gold accurately. Considering the result, they conclude that the random forest algorithm is better than other algorithms [1]. In this research, they indicated that the most important factors affecting the price of gold were the USD/INR exchange rate, the S&P 500 index, and the crude oil price. They used supervised learning algorithms such as Random Forest Regressor for high accuracy in predicting output [2]. This research was carried out for data between April 2020 and January 2022 on oil worth, NYSE, normal and Poor's (S&P) five hundred indexes, US bond rates (10 years), and EuroUSD exchange rates. They conclude that random forest machine learning beats customary and current prediction models [3]. In this research paper, the authors used three machine learning algorithms: linear regression, random forest regression, and analysis. Models were used to analyze data. They studied the relationship between gold prices, selected economies, and various market variables to try and accurately predict the future price of gold using machine learning algorithms [4]. In this research, they developed a machine that predicts prices using historical data. It'll help investors invest in gold at the right time to reap benefits and mitigate their losses. They used price regression prices to predict the prices. The authors used data

from prices from 2000 to 2024 on gold prices, interest rates, inflation rates, GDP, oil prices, and exchange rates [5]. The Dollar Index U., the Federal Funds Rate U., the Consumer Price Index (CPI), the exchange rate of the dollar to the Chinese yuan, the price of oil, and the S&P 500 index using these variables predicted gold prices. In this research, various machine learning algorithms were employed, including linear regression, decision tree regression, and random forest regression. They conclude that the random forest algorithm provides more accurate results [6]. In this paper author explores the use of machine learning algorithms to predict gold prices based on historical data and their correlations to develop predictive models. They used algorithms such as Linear Regression and Decision Tree. They conclude that the Decision Tree accurately identified complex patterns in gold price data [7].

III. METHODOLOGY



Figure 1: Machine Learning Model Workflow

1) Data Preprocessing:

1) Loading dataset: Firstly, we saved the dataset in a CSV file and loaded it into Python using the Pandas library. We used the `read_csv()` function to open the CSV file, to read the data, and to convert the data into table format for easy analysis.

2) Data exploration: To understand the structure of the dataset, we displayed the first and last five rows as well as used the `shape` command to check the number of rows and columns.

3) Handling Missing Values: We used the `isnull()` function to check for missing values in the dataset. There were no missing values; hence, the dataset is complete and reliable for analysis.

2) Data Collection:

We build a dataset using historical data from Investing.com a popular website from January 2015 until February 2025. There are 122 monthly entries in this dataset, with each entry containing a date along with some financial data. Let's take a look at the Features of the dataset.

1. Date: This column indicates the date of every entry. This helps observe changes in gold prices over time.

2. SPX (S&P 500 Index): It shows the movement of the stock market of the United States. It gives the performance of 500 of the largest traded companies in the United States. It has a negative correlation with gold prices.

3. GLD (Gold Price): This column indicates the market price of gold in the United States, which is utilized as the target variable in the forecast. Gold prices are affected by oil prices, the performance of the stock market, economic conditions, geopolitical issues, and demand & supply.

4. USO (Crude Oil Price): It shows the price of oil in the US. Oil prices and gold prices go in opposite directions based on the economy, but they can move in the same direction due to inflation or a weak currency.

5. SLV (Silver Price): This column shows the price of silver in the United States. It is considered a valuable metal. Its price moves together with gold because of the U.S. dollar, economic conditions and inflation.

6. EUR/USD (Exchange Rate): The EUR/USD exchange rate indicates the value of one Euro in terms of U.S. dollars.

3) Exploratory Data Analysis

1) Statistical Summary: We used the `gold_data.describe()` function, which gives a statistical summary of the data like mean, standard deviation (std), minimum (min), maximum (max), and quartiles (25%, 50%, 75%), which helps to understand the data distribution.

2) Distribution of gold prices:

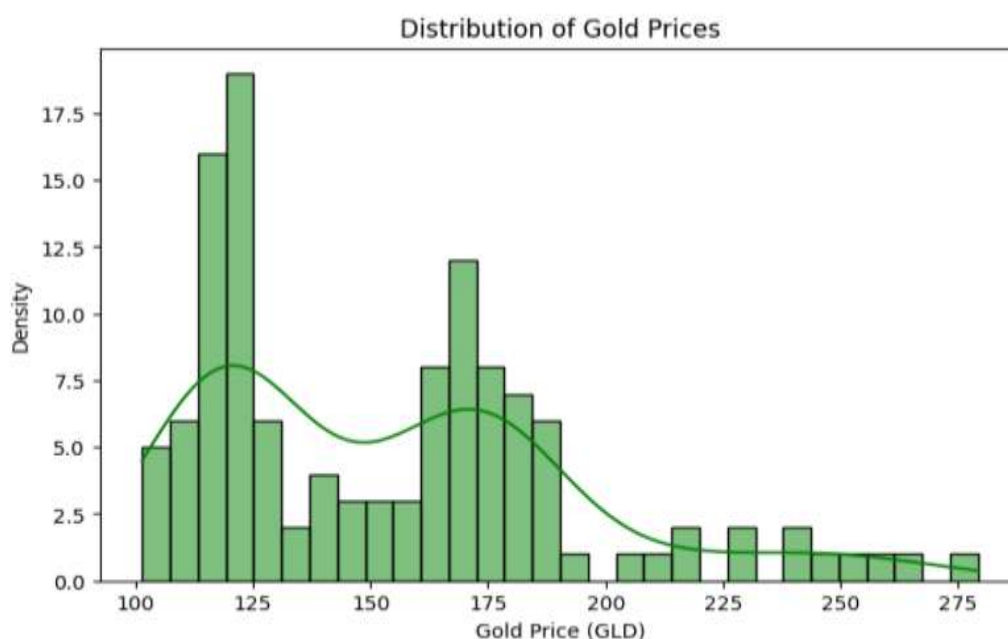


Figure 2: Distribution of Gold Prices

Fig 2. shows the distribution of gold prices (GLD). In the graph, most gold prices are between 100 and 175, while higher prices above 200 are less frequent. The green line shows the overall pattern of price changes.

3) Feature Importance

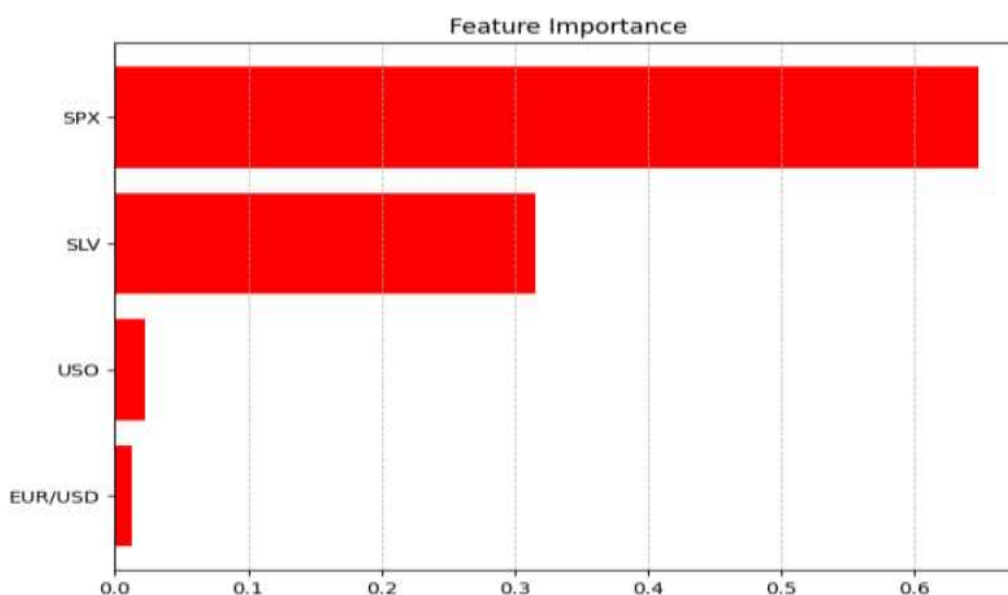


Figure 3: Feature Importance

Fig. 3 displays the most important features in predicting gold prices. The graph indicates that SPX (Stock Market) and SLV (Silver Prices) are the most important factors that impact gold prices. However, oil prices (USO) and the Euro/USD exchange rate have only a minor effect.

4) Gold Price Trend:

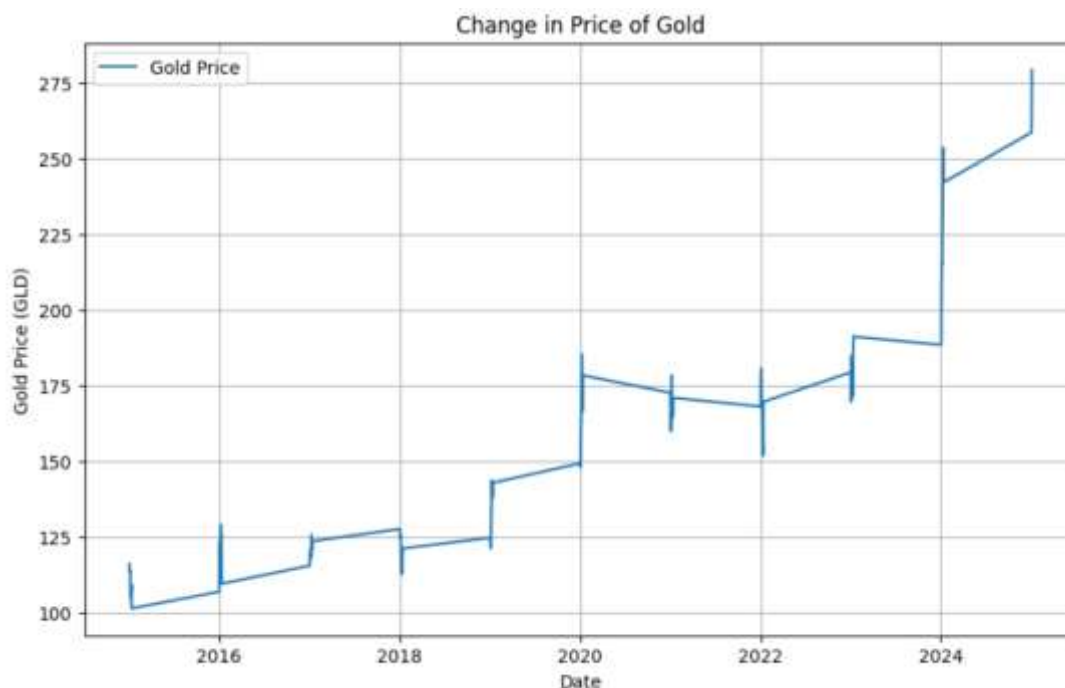


Figure 4: Gold Price Trend

Fig 4. shows the gold price trend from 2015 to 2025. The x-axis represents the date, while the y-axis represents the gold price. The graph shows that the Gold prices (GLD) have mostly increased over the years. Around 2020–2022, the price slightly decreased but after 2023, prices increased quickly, showing strong growth in gold prices. Overall, the trend suggests that gold prices will keep increasing in the future.

4) Model Selection

In this study we used regressor models such as Random Forest, Decision Trees, Linear Regression, Support Vector Machine (SVM), and K-Nearest Neighbours (KNN), to predict future gold prices.

- **Random Forest Regressor:** Random Forest Regressor is a machine learning algorithm that makes predictions using a large number of decision trees. It trains multiple trees on randomly chosen subsets of data and takes the average of their outputs to improve accuracy. This method reduces overfitting and performs well with both linear and nonlinear relationships. Averaging the forecasts of many trees reduces errors and finds complex patterns, thus making it a good forecasting method.
- **Decision Trees:** A Decision Tree Regressor is a machine learning algorithm that makes predictions by dividing data into branches based on conditions. It learns from historical data to find patterns and make future predictions. It can sometimes be too detailed, so combining many trees (like in Random Forest) it makes predictions more accurate.
- **Linear Regression:** Linear Regression is a supervised machine learning algorithm used to find the relationship between feature variables (independent) and the target variable (dependent), and that relationship can be represented by a straight line. It also finds the errors between the predicted and actual values. Linear regression helps us understand how the variations in input features affect the output, and it uses this relationship to make predictions on new data.
- **Support Vector Machine (SVM):** Support Vector Machine Regression (SVR) is a machine learning algorithm used to predict continuous values and find the best-fit line allowing small errors. It handles both linear and non-linear(complex) relationships in the data. However, it can be slow with large datasets and requires careful selection of parameters for the best results.

- **K-Nearest Neighbours (KNN):** K-Nearest Neighbours (KNN) Regression is a supervised learning algorithm that predicts continuous numeric values. It identifies K nearest points to an input based on Euclidean distance. KNN is easy to understand, it just stores the data and makes predictions when required. A small K makes the prediction highly sensitive to near points, whereas a large K generates smoother results but may lose precision. KNN works well with small sets and simple patterns.

5) Model Training:

We trained different machine learning models to predict gold prices using features like SPX, USO, SLV, and EUR/USD. We split the dataset into 80% training data and 20% testing data. Different models were tested like Linear Regression, KNN, Decision Tree, Random Forest, and SVR. They were evaluated using error metrics such as MAE, MSE, RMSE, and R^2 Score.

IV. RESULT

Model	Mean Absolute Error	Mean Squared Error	Root Mean Squared Error	R-squared(R^2)
Linear Regression	6.045218	81.668881	9.037084	0.952000
K-Nearest Neighbours Regressor	7.038800	79.321990	8.906289	0.953379
Decision Tree Regressor	6.418800	100.344540	10.017212	0.941023
Random Forest Regressor	5.454324	63.802124	7.987623	0.962501
Support Vector Regressor (SVR)	23.490232	901.011541	30.016854	0.470435

Figure 5: Model Evaluation Results

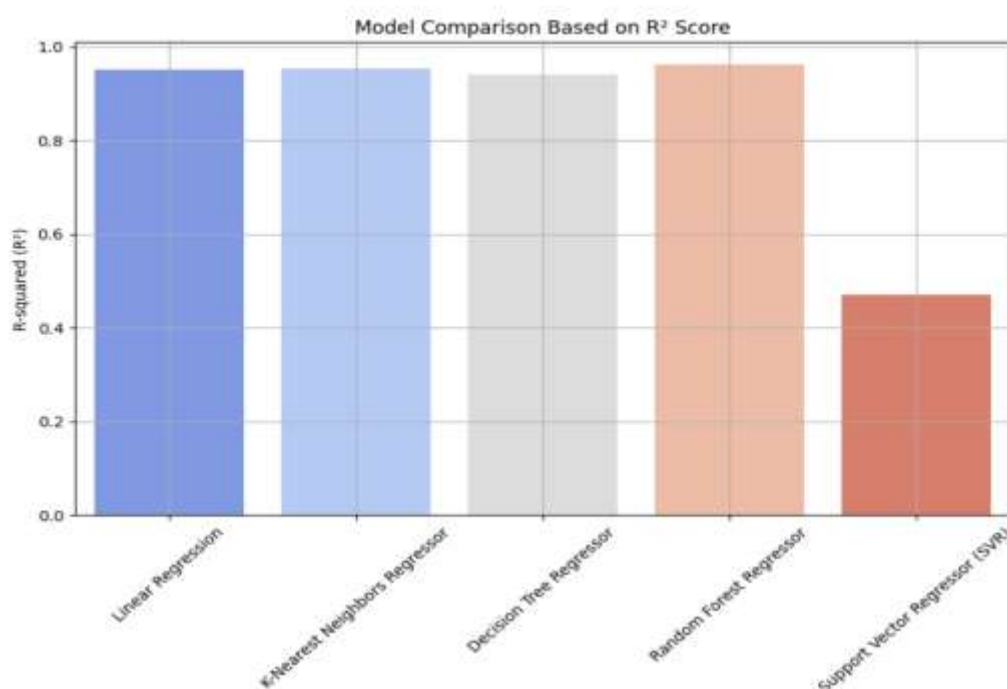


Figure 6: Model Comparison

- Fig. 5 displays that the Random Forest Regressor performed best with the lowest MAE (5.45), MSE (63.80), and RMSE (7.98) and highest R-squared (0.9625).
- K-Nearest Neighbors (KNN) Regressor and Linear Regression also performed well with R^2 values of 0.9534 and 0.9520 but with slightly higher error values than the Random Forest Regressor.
- Decision Tree Regressor gave a slightly lower R^2 score (0.9410) and a higher RMSE (10.01).
- Support Vector Regressor (SVR) gave the worst result with the highest MAE (23.49), RMSE (30.02), and the lowest R-squared (0.4704). This means the SVR is the worst model for gold price prediction.
- Hence, the Random Forest Regressor is the best model for predicting gold prices because it has the highest accuracy and the lowest errors.

V. MODEL EVALUATION

The results show that the random forest regressor predicts gold prices accurately. Therefore, we used the random forest algorithm for model evaluation.

Model Evaluation

```
# prediction on Test Data
best_model = RandomForestRegressor(n_estimators=100, random_state=2)
best_model.fit(X_train, y_train)
test_data_prediction = best_model.predict(X_test)

print(test_data_prediction)

[125.1355 125.8043 122.0841 166.3186 111.7949 114.9399 135.1992 189.0458
 121.171 138.5012 136.9696 251.1485 117.2238 153.6266 206.6604 114.2789
 168.4811 147.0943 258.3287 113.3155 119.7782 120.8639 188.9982 180.1367
 114.8287]
```

Figure 7: Model Evaluation

- **Comparison of Actual and Predicted Values:**



Figure 8: Actual Price and Predicted Price

In Fig. 7, using a line graph, the actual gold prices (green line) and predicted prices (red dashed line) are compared. The graph shows that the two lines are closely aligned, which means that the model predicts prices accurately with only small differences. According to the graph, the model successfully captures trends and movements in gold prices.

VI. CONCLUSION

This report aims to identify the best machine learning model that accurately forecasts gold prices using the provided dataset. The report follows a step-by-step approach, beginning with gathering and cleaning the data and then selecting machine learning algorithms. We analyzed patterns using charts and tested various prediction models to evaluate their accuracy. According to the result, the random forest model gives the highest R2 score (0.9625) compared to the other models. Therefore, the Random Forest Regressor is the most suitable model for gold price prediction.

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